Supplementary Materials for: Toward a Better NSGA-III: Addressing Consistency, Stability, and Uniformity

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Appendix

S1 Ablation Study Results

We discuss interesting results:

On MOTSP ($m = 3$), Tk-NSGA-III is slower than NSGA-III because one of the heuristic solutions (the center one) is the extreme solution for all three reference vectors, causing most of NSGA-III generations to "skip" the hyperplane system because the system is singular. It falls back to using the worst-of-front or worst-of-population (depending on the situation) most of the time, causing the time to be faster. It can be seen that in this case, Tk-NSGA-III offers a definitive boost in performance.

For the specific instance of five-objective MOTSP we tested, the Dss-NSGA-III outperforms NSGA-III$^+$ consistently. If we looked at the three individual components: Tk adds performance, Dss adds performance, but ZYX lowers performance. The only reasonable explanation is that because the corner solutions are preserved, it may have somehow slowed down the convergence. Another thing to notice is that none of the algorithms nearly converged toward the Pareto front (judging by the reference HV), so this may be a particularly difficult instance, and usually we cannot conclude anything significant on such a problem. Overall, we are not entirely sure what happened with this problem, but we will be leaving the investigation of this to future work.

Table 1. Ablation Study Results (Part 1 of 2)

Problem	Algorithm	HV (↑)	IGD$^+$ (↓)	Δ^* (↓)	Time (↓)
	NSGA-III	0.6389 ± 0.0000	2.09e-03 ± 4.73e-07	0.9680 ± 0.0001	15.00 ± 0.65
	ZYX-NSGA-III	0.6385 ± 0.0001	2.26e-03 ± 8.47e-05	1.0142 ± 0.0958	16.06 ± 0.90
RWA1	Tk-NSGA-III	**0.6389** ± 0.0000	**2.09e-03** ± 5.52e-07	**0.9680** ± 0.0001	**14.81** ± 0.75
Ref. HV: 0.6427	Dss-NSGA-III	0.6389 ± 0.0000	2.09e-03 ± 5.69e-07	0.9680 ± 0.0001	19.99 ± 0.86
	ZYX-Tk-NSGA-III	0.6385 ± 0.0001	2.25e-03 ± 5.88e-05	1.0153 ± 0.0795	16.17 ± 0.66
	ZYX-Tk-Dss-NSGA-III	0.6386 ± 0.0001	2.24e-03 ± 4.96e-05	1.0017 ± 0.0681	20.54 ± 1.14
	NSGA-III	0.9386 ± 0.0021	1.36e-02 ± 7.57e-04	1.3412 ± 0.2388	19.49 ± 0.52
	ZYX-NSGA-III	0.9372 ± 0.0052	1.45e-02 ± 3.32e-03	1.3039 ± 0.2631	22.60 ± 1.33
RWA2	Tk-NSGA-III	0.9402 ± 0.0017	1.18e-02 ± 4.73e-04	1.2835 ± 0.2402	**16.74** ± 0.54
Ref. HV: 0.9608	Dss-NSGA-III	0.9442 ± 0.0068	9.52e-03 ± 3.89e-03	**0.9863** ± 0.4782	26.02 ± 0.94
	ZYX-Tk-NSGA-III	0.9417 ± 0.0019	1.19e-02 ± 4.12e-04	1.3843 ± 0.2327	19.91 ± 0.99
	ZYX-Tk-Dss-NSGA-III	**0.9449** ± 0.0023	**8.70e-03** ± 5.08e-04	1.0987 ± 0.5285	25.44 ± 1.42
	NSGA-III	0.7529 ± 0.0032	2.85e-02 ± 6.02e-04	0.7533 ± 0.0158	**20.25** ± 1.20
	ZYX-NSGA-III	0.7516 ± 0.0028	2.87e-02 ± 3.98e-04	0.7415 ± 0.0122	22.96 ± 1.95
RWA3	Tk-NSGA-III	0.7523 ± 0.0032	2.86e-02 ± 4.84e-04	0.7518 ± 0.0113	20.73 ± 0.92
Ref. HV: 0.8077	Dss-NSGA-III	**0.7630** ± 0.0008	**2.58e-02** ± 2.16e-04	**0.6015** ± 0.0042	25.38 ± 0.96
	ZYX-Tk-NSGA-III	0.7504 ± 0.0027	2.88e-02 ± 4.43e-04	0.7419 ± 0.0130	23.46 ± 1.24
	ZYX-Tk-Dss-NSGA-III	0.7621 ± 0.0008	2.62e-02 ± 2.46e-04	0.6099 ± 0.0026	26.73 ± 1.46
	NSGA-III	1.2057 ± 0.0013	2.31e-03 ± 4.47e-04	1.2160 ± 0.0731	**18.77** ± 0.63
	ZYX-NSGA-III	1.2078 ± 0.0004	1.90e-03 ± 1.90e-04	1.2728 ± 0.0770	21.76 ± 1.22
RWA4	Tk-NSGA-III	1.2061 ± 0.0014	2.08e-03 ± 3.96e-04	1.2518 ± 0.0955	18.80 ± 0.72
Ref. HV: 1.2119	Dss-NSGA-III	1.2105 ± 0.0002	3.46e-04 ± 5.06e-05	**0.8544** ± 0.0363	25.28 ± 1.00
	ZYX-Tk-NSGA-III	1.2077 ± 0.0004	1.92e-03 ± 1.80e-04	1.2828 ± 0.0603	22.15 ± 1.02
	ZYX-Tk-Dss-NSGA-III	**1.2110** ± 0.0001	**3.43e-04** ± 3.57e-05	0.8593 ± 0.0221	27.86 ± 1.63
	NSGA-III	0.5544 ± 0.0020	9.19e-03 ± 1.12e-03	1.1628 ± 0.0646	**19.20** ± 0.72
	ZYX-NSGA-III	0.5554 ± 0.0012	9.48e-03 ± 1.05e-03	1.1612 ± 0.0740	22.95 ± 1.08
RWA5	Tk-NSGA-III	0.5536 ± 0.0021	9.92e-03 ± 1.38e-03	1.2169 ± 0.0875	19.43 ± 1.14
Ref. HV: 0.5702	Dss-NSGA-III	0.5601 ± 0.0007	5.09e-03 ± 2.82e-04	0.8868 ± 0.0928	25.82 ± 0.92
	ZYX-Tk-NSGA-III	0.5551 ± 0.0012	9.79e-03 ± 1.12e-03	1.1907 ± 0.0732	23.15 ± 1.03
	ZYX-Tk-Dss-NSGA-III	**0.5614** ± 0.0004	**5.05e-03** ± 2.42e-04	**0.7953** ± 0.0184	28.94 ± 1.68
	NSGA-III	0.4142 ± 0.0020	1.59e-02 ± 2.29e-03	0.8974 ± 0.0859	16.43 ± 0.84
	ZYX-NSGA-III	0.4141 ± 0.0018	1.66e-02 ± 2.20e-03	0.8937 ± 0.0861	19.20 ± 1.50
RWA6	Tk-NSGA-III	0.4148 ± 0.0015	1.56e-02 ± 1.12e-03	0.8831 ± 0.0309	**16.33** ± 1.04
Ref. HV: 0.4314	Dss-NSGA-III	**0.4187** ± 0.0006	**9.21e-03** ± 4.80e-04	0.6815 ± 0.0193	23.08 ± 0.85
	ZYX-Tk-NSGA-III	0.4148 ± 0.0014	1.65e-02 ± 1.51e-03	0.8847 ± 0.0451	19.32 ± 1.13
	ZYX-Tk-Dss-NSGA-III	0.4183 ± 0.0009	9.44e-03 ± 4.43e-04	**0.6637** ± 0.0162	25.30 ± 1.74

Table 2. Ablation Study Results (Part 2 of 2)

Problem	Algorithm	HV (↑)	IGD$^+$ (↓)	Δ^* (↓)	Time (↓)
RWA7	NSGA-III	0.7841 ± 0.0024	2.24e-02 ± 1.20e-03	0.8965 ± 0.0504	**17.14** ± 0.60
	ZYX-NSGA-III	0.7866 ± 0.0015	2.28e-02 ± 1.12e-03	0.8933 ± 0.0421	19.94 ± 0.91
Ref. HV: 0.8220	Tk-NSGA-III	0.7841 ± 0.0025	2.23e-02 ± 1.32e-03	0.8893 ± 0.0468	17.22 ± 0.56
	Dss-NSGA-III	0.7897 ± 0.0013	**1.71e-02** ± 4.90e-04	**0.6009** ± 0.0095	23.27 ± 0.74
	ZYX-Tk-NSGA-III	0.7866 ± 0.0018	2.28e-02 ± 1.26e-03	0.8899 ± 0.0456	20.06 ± 0.94
	ZYX-Tk-Dss-NSGA-III	**0.7935** ± 0.0008	1.74e-02 ± 3.74e-04	0.6038 ± 0.0131	25.43 ± 1.50
RWA8	NSGA-III	0.9701 ± 0.0044	3.22e-02 ± 1.99e-03	0.7474 ± 0.0423	19.90 ± 0.70
	ZYX-NSGA-III	0.9713 ± 0.0052	3.36e-02 ± 2.69e-03	0.7142 ± 0.0416	25.06 ± 1.45
Ref. HV: 1.0387	Tk-NSGA-III	0.9692 ± 0.0048	3.20e-02 ± 2.18e-03	0.7416 ± 0.0515	**19.76** ± 0.58
	Dss-NSGA-III	0.9840 ± 0.0029	**2.30e-02** ± 9.17e-04	0.5326 ± 0.0189	26.84 ± 0.99
	ZYX-Tk-NSGA-III	0.9715 ± 0.0056	3.34e-02 ± 3.03e-03	0.7298 ± 0.0413	24.58 ± 1.59
	ZYX-Tk-Dss-NSGA-III	**0.9890** ± 0.0027	2.34e-02 ± 1.10e-03	**0.5324** ± 0.0175	29.49 ± 2.24
RWA9	NSGA-III	1.6511 ± 0.0055	1.07e-02 ± 1.03e-03	0.9349 ± 0.0764	45.94 ± 2.65
	ZYX-NSGA-III	**1.6620** ± 0.0031	**1.05e-02** ± 5.30e-04	0.8644 ± 0.0650	46.84 ± 4.14
Ref. HV: 1.6873	Tk-NSGA-III	1.6461 ± 0.0077	1.14e-02 ± 1.07e-03	0.9641 ± 0.1276	**36.38** ± 3.36
	Dss-NSGA-III	1.6445 ± 0.0051	1.36e-02 ± 4.96e-04	**0.5045** ± 0.0067	49.25 ± 2.80
	ZYX-Tk-NSGA-III	1.6609 ± 0.0048	1.11e-02 ± 1.34e-03	0.9037 ± 0.0841	39.47 ± 3.32
	ZYX-Tk-Dss-NSGA-III	1.6538 ± 0.0045	1.29e-02 ± 6.69e-04	0.5086 ± 0.0103	45.66 ± 3.50
RWA10	NSGA-III	1.6956 ± 0.0267	5.61e-02 ± 3.58e-03	0.7037 ± 0.0209	47.02 ± 2.51
	ZYX-NSGA-III	1.7750 ± 0.0150	5.51e-02 ± 2.26e-03	0.6947 ± 0.0264	49.98 ± 3.61
Ref. HV: 2.0341	Tk-NSGA-III	1.6980 ± 0.0345	5.60e-02 ± 3.15e-03	0.7114 ± 0.0322	**40.94** ± 2.33
	Dss-NSGA-III	1.8176 ± 0.0082	**4.19e-02** ± 8.11e-04	**0.4904** ± 0.0064	51.79 ± 3.21
	ZYX-Tk-NSGA-III	1.7752 ± 0.0123	5.52e-02 ± 1.88e-03	0.6883 ± 0.0234	44.00 ± 3.41
	ZYX-Tk-Dss-NSGA-III	**1.8280** ± 0.0078	4.27e-02 ± 8.13e-04	0.4952 ± 0.0081	50.13 ± 3.45
MOTSP ($m = 3$)	NSGA-III	0.6772 ± 0.0152	5.98e-02 ± 6.35e-03	0.7172 ± 0.1281	**63.08** ± 18.17
	ZYX-NSGA-III	0.6770 ± 0.0150	6.00e-02 ± 7.73e-03	0.6648 ± 0.0601	64.57 ± 19.75
Ref. HV: 0.8086	Tk-NSGA-III	0.6898 ± 0.0111	5.42e-02 ± 5.71e-03	0.7219 ± 0.1212	81.24 ± 2.57
	Dss-NSGA-III	0.7008 ± 0.0142	5.29e-02 ± 6.73e-03	0.6982 ± 0.0814	74.63 ± 11.66
	ZYX-Tk-NSGA-III	0.6831 ± 0.0145	5.78e-02 ± 5.86e-03	0.6498 ± 0.0703	84.50 ± 3.05
	ZYX-Tk-Dss-NSGA-III	**0.7063** ± 0.0100	**5.14e-02** ± 5.34e-03	**0.6162** ± 0.0622	88.43 ± 3.08
MOTSP ($m = 5$)	NSGA-III	0.3305 ± 0.0163	3.71e-01 ± 2.09e-02	0.2946 ± 0.0215	**158.77** ± 51.79
	ZYX-NSGA-III	0.3243 ± 0.0249	3.81e-01 ± 4.29e-02	0.3175 ± 0.0223	162.38 ± 53.44
Ref. HV: 0.8765	Tk-NSGA-III	0.3465 ± 0.0270	3.59e-01 ± 3.07e-02	0.3013 ± 0.0236	203.00 ± 7.22
	Dss-NSGA-III	**0.4203** ± 0.0230	**2.47e-01** ± 2.52e-02	**0.2779** ± 0.0307	180.04 ± 31.87
	ZYX-Tk-NSGA-III	0.3458 ± 0.0214	3.70e-01 ± 2.65e-02	0.3156 ± 0.0220	207.65 ± 5.57
	ZYX-Tk-Dss-NSGA-III	0.3979 ± 0.0277	2.83e-01 ± 3.12e-02	0.2787 ± 0.0251	212.82 ± 5.75
MOTSP ($m = 10$)	NSGA-III	0.7174 ± 0.0368	1.93e-01 ± 1.35e-02	0.7572 ± 0.0703	402.33 ± 14.34
	ZYX-NSGA-III	0.8225 ± 0.0219	1.63e-01 ± 1.16e-02	0.4171 ± 0.0459	407.96 ± 14.28
Ref. HV: 1.2334	Tk-NSGA-III	0.7283 ± 0.0442	1.88e-01 ± 1.42e-02	0.7921 ± 0.0882	**397.07** ± 13.02
	Dss-NSGA-III	0.7690 ± 0.0266	1.80e-01 ± 1.21e-02	0.6405 ± 0.0702	407.26 ± 15.97
	ZYX-Tk-NSGA-III	0.8548 ± 0.0427	1.54e-01 ± 1.26e-02	0.4491 ± 0.0578	401.82 ± 12.82
	ZYX-Tk-Dss-NSGA-III	**0.8699** ± 0.0253	**1.51e-01** ± 9.84e-03	**0.3749** ± 0.0374	405.46 ± 12.58

S2 Derivation of Tikhonov Regularization Bound

To establish the bound on the temporal difference $\|p_{t+1} - p_t\|$, we first define the regularized matrix and the target vector for a given generation t as:

$$M_t = A_t^\top A_t + \lambda I, \quad y_t = A_t^\top \mathbf{1}^m + \lambda p^*. \tag{1}$$

The closed-form solution is thus $p_t = M_t^{-1} y_t$. The difference between the estimated hyperplane parameters of two successive generations is:

$$p_{t+1} - p_t = M_{t+1}^{-1} y_{t+1} - M_t^{-1} y_t. \tag{2}$$

Adding and subtracting $M_{t+1}^{-1} y_t$ isolates the changes in the vector from the changes in the matrix:

$$p_{t+1} - p_t = M_{t+1}^{-1}(y_{t+1} - y_t) + (M_{t+1}^{-1} - M_t^{-1}) y_t. \tag{3}$$

We apply the matrix inverse identity $X^{-1} - Y^{-1} = X^{-1}(Y - X)Y^{-1}$ to the second term:

$$M_{t+1}^{-1} - M_t^{-1} = M_{t+1}^{-1}(M_t - M_{t+1}) M_t^{-1}. \tag{4}$$

Substituting this back and recognizing that $M_t^{-1} y_t = p_t$, we obtain:

$$\begin{aligned} p_{t+1} - p_t &= M_{t+1}^{-1}(y_{t+1} - y_t) + M_{t+1}^{-1}(M_t - M_{t+1}) p_t \\ &= M_{t+1}^{-1}\big[(y_{t+1} - y_t) - (M_{t+1} - M_t) p_t\big]. \end{aligned} \tag{5}$$

Expanding the definitions of M_t and y_t, we note that the constant regularization terms λp^* and λI cancel out exactly during subtraction:

$$y_{t+1} - y_t = A_{t+1}^\top \mathbf{1}^m - A_t^\top \mathbf{1}^m, \tag{6}$$

$$M_{t+1} - M_t = A_{t+1}^\top A_{t+1} - A_t^\top A_t. \tag{7}$$

This yields the exact difference equation:

$$p_{t+1} - p_t = (A_{t+1}^\top A_{t+1} + \lambda I)^{-1} \Big[(A_{t+1}^\top \mathbf{1}^m - A_t^\top \mathbf{1}^m) - (A_{t+1}^\top A_{t+1} - A_t^\top A_t) p_t\Big]. \tag{8}$$

Taking the L_2 norm of both sides, and applying the triangle inequality along with the submultiplicative property of matrix norms ($\|AB\| \leq \|A\|\|B\|$), we get:

$$\|p_{t+1} - p_t\| \leq \|(A_{t+1}^\top A_{t+1} + \lambda I)^{-1}\| \Big(\|A_{t+1}^\top \mathbf{1}^m - A_t^\top \mathbf{1}^m\| + \|A_{t+1}^\top A_{t+1} - A_t^\top A_t\| \, \|p_t\|\Big). \tag{9}$$

Since the eigenvalues of the positive definite matrix $A_{t+1}^\top A_{t+1} + \lambda I$ are strictly bounded below by $\lambda > 0$, its inverse norm is bounded by $1/\lambda$. This leads to the final stability bound:

$$\|p_{t+1} - p_t\| \leq \frac{1}{\lambda} \Big(\|A_{t+1}^\top \mathbf{1}^m - A_t^\top \mathbf{1}^m\| + \|A_{t+1}^\top A_{t+1} - A_t^\top A_t\| \, \|p_t\|\Big). \tag{10}$$